



SUNDAR RANGACHARI

AI-Augmented Data Architect & Lead Database Developer

AI-assisted engineering · LLM-driven development · Oracle PL/SQL · Data platforms for AI

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Executive profile

Lead database developer and hands-on Oracle architect with 25+ years turning complex, high-volume data estates into dependable platforms across healthcare payment integrity, banking, telecom billing, rail, energy and supply chain. Works AI-augmented: Cursor AI and Claude AI sit in the daily development loop for code generation, refactoring, legacy comprehension and LLM-assisted review, with every generated artifact reviewed, tested and tuned before production. The same discipline shapes the data itself, building normalized, governed, lineage-tracked estates that AI and analytics systems can actually consume. Individual contributor who leads: architecture first, performance always, human in the loop.

Signature outcomes

- **AI-augmented engineering.** Brought Cursor AI and Claude AI into daily PL/SQL development, legacy code review and documentation, shortening build and refactor cycles and raising code consistency across the team.
- **A platform built from zero.** Designed and delivered the Enterprise Content Library at Cotiviti, a custom Oracle platform that unified content from disparate legacy systems into one governed source, with the normalized data model, ERD and ETL owned personally.
- **Performance as a discipline.** Restored throughput on business-critical Oracle workloads through bulk collect redesign, partitioning, defragmentation and explain plan and TKPROF tuning, converting unpredictable long-running batch into runs the business can schedule around.
- **Modernization without disruption.** Carried VMS-era applications, Oracle Forms 6i estates and character-based systems onto modern Unix, Forms 10g and web platforms for British Petroleum, MIT and RGIS with service continuity throughout.
- **Teams that outlast the engagement.** Architected onsite and offshore delivery models and mentored technical teams for clients across the US, UK, Canada, India and the Middle East, embedding code review, reusable component libraries and deployment standards as the default way of working.

AI engineering practice

AI-assisted development

Cursor AI and Claude AI in the daily loop for PL/SQL generation, refactoring and legacy comprehension, with prompt patterns tuned to Oracle schema and package structure.

LLM-assisted code review

Large legacy PL/SQL estates swept with LLM review to surface dead code, unsafe SQL, missing bind variables and tuning candidates before a human reviewer spends time on them.

Analytics and ML data pipelines

ETL that produces clean, versioned, well-typed datasets fit for downstream analytics and machine learning consumption rather than one-off extracts.

Prompt engineering for data work

Reusable prompt patterns for schema explanation, query optimization, migration mapping and specification-to-DDL translation.

Agentic coding workflows

Multi-step, tool-using AI sessions driving repetitive database work: scaffolding packages, generating test data, drafting unit tests and producing technical documentation from schema.

Retrieval-ready data curation

Normalized, deduplicated, lineage-tracked content models of the kind retrieval-augmented generation depends on, assembled from disparate legacy sources.

Data governance for AI

Access control, integrity constraints, audit trails and documented data contracts, so any model or agent draws on data with known provenance.

Human in the loop

AI output treated as a first draft, never a deliverable: reviewed, unit tested and performance tuned before it reaches a production database.

Cursor AI Claude AI LLM-assisted development Agentic workflows Prompt engineering AI code review
Retrieval-ready data Data lineage Feature-ready datasets Human in the loop AI data governance
Legacy comprehension

Core capabilities

AI-assisted engineering

Cursor AI and Claude AI for PL/SQL generation, refactoring, legacy code comprehension, test data and documentation; curation and modeling of data estates for AI and analytics consumption

Data architecture & modeling

Logical and physical design, normalization, ERD, SQL Data Modeler, ERWIN, DFD, HLD and LLD, data migration strategy

ETL & integration

Extraction, transformation and loading, data warehousing and OLTP concepts, integration with SAP, JD Edwards, Salesforce, BIOVIA and Microsoft portals

Platforms & automation

Advanced Unix and Korn shell scripting, Linux, Solaris, HP-UX, AIX, Open VMS, Windows Server, cron and Windows Task Scheduler automation, CA scheduler

Delivery leadership

Team lead and mentoring, code review culture, estimation, RFP and costing, migration and deployment strategy, SDLC documentation, client-facing liaison

Oracle development

Advanced PL/SQL, packages, procedures, functions and triggers, bulk collect and bulk processing, Pro*C, Oracle Forms and Reports 6i and 10g, SQL*Loader, SQL*Plus

Performance engineering

Explain plan, optimizer hints, TKPROF, AWR and ADDM analysis, SQL Tuning Advisor, partitioning, defragmentation, memory, I/O and disk tuning

Database administration

Oracle 7.x through 19c, RMAN, Export/Import and Data Pump, Streams replication, standby databases, Flashback, cloning and refresh, patch management, capacity planning, ASM and SAN

Reporting & BI

OBIEE, BI Publisher, SSRS, Discoverer Plus, Analytic Workspace Manager, Crystal Reports

Professional experience

Cotiviti Atlanta, GA

Jul 2024 - Present

Lead Database Developer & DBA · Oracle PL/SQL and DBA, 12c and 19c

Own development and administration of the Oracle estate behind the Enterprise Content Library, from data model through ETL, tuning and resilience.

- **AI in the daily loop.** Applied Cursor AI and Claude AI to PL/SQL authoring, legacy package comprehension and review, cutting the time between requirement and working code and making unfamiliar legacy logic accessible to the whole team.
- **Platform delivered end to end.** Built the Enterprise Content Library as a complete custom project and integrated it with the surrounding legacy systems, giving the business a single governed content source in place of scattered stores.
- **Model that holds up.** Authored the normalized data model and ERD for the entire project, so downstream ETL and reporting extend the design rather than work around it.
- **Consolidated ingestion.** Designed the ETL that pulls content into the library from multiple source systems, replacing point-to-point extracts with a repeatable, monitored flow.
- **Throughput restored.** Re-engineered complex procedures, functions and packages with bulk collect and bulk processing, turning heavy SQL and PL/SQL workloads into predictable runs.
- **Recoverable by design.** Set backup and recovery strategy across RMAN, Data Pump and Streams replication with standby databases, cloning, refresh and patch management, keeping restore a rehearsed procedure rather than an incident.
- **Hands off the routine.** Automated recurring database and file operations through advanced shell scripting, removing manual steps from the operations calendar.

Merck Animal Health (through HCL) Atlanta, GA

Feb 2021 - Jun 2024

Lead Database Developer & DBA · HomeAgain platform, Oracle 12c and 19c

Database owner for the HomeAgain application, responsible for migration off legacy systems, performance recovery and integration with commercial systems.

- **Legacy retired cleanly.** Migrated data from legacy systems into the HomeAgain application as database administrator, consolidating the operational record onto one supported platform.
- **Performance reclaimed.** Tuned existing legacy PL/SQL and partnered with the hosting vendor, Connectria, on database-level tuning, lifting responsiveness on the workloads the business felt most.
- **Commercial systems connected.** Designed the ETL extracting operational data into Salesforce, putting current customer and product data in front of commercial teams.
- **Reusable custom layer.** Built custom PL/SQL components for the legacy estate, replacing one-off fixes with packaged, supportable logic.
- **Bulk processing at scale.** Applied bulk collect and set-based rewrites to the heaviest SQL and PL/SQL paths, keeping batch inside its window as volumes grew.
- **Operations automated.** Scripted routine database and interface tasks in advanced shell, reducing manual intervention in day-to-day running.

Cotiviti (through UST Global) Atlanta, GA

May 2019 - Feb 2021

Lead Database Developer & DBA · Enterprise Content Library, Oracle 12c and 19c

Established the Enterprise Content Library from a blank page: data model, PL/SQL layer, ETL and the administration practice around it.

- **From concept to running system.** Designed and built the Enterprise Content Library as a complete custom project and integrated it with other legacy systems, creating the platform the later engagement would extend.
- **Foundation set correctly.** Produced the normalization and ERD for the whole project and stood up database creation, maintenance and data modeling practice alongside it.
- **Complex logic packaged.** Developed the complex procedures, functions and packages carrying the library's business rules, keeping logic in one reviewable place.
- **Sources unified.** Designed the ETL drawing content into the library from a range of upstream systems.
- **Tuned for the workload.** Applied Oracle performance tuning and bulk processing across SQL and PL/SQL so ingestion and retrieval held up under production volume.
- **Continuity assured.** Owned backup and recovery strategy, capacity planning, replication, standby databases, cloning, refresh and patching for the environment.

RGIS Auburn Hills, MI

Apr 2018 - May 2019

Lead Technical Consultant · Billing Portfolio System, Oracle 11g, 12c and 18c

Delivered a custom billing platform into Oracle EBS and modernized the Forms estate behind it.

- **Billing platform delivered.** Built the Billing Portfolio System as a complete custom project and integrated it into Oracle EBS, with the complex procedures, functions and packages behind it.
- **Forms estate modernized.** Migrated Oracle Forms 6i to Forms 10g and customized the resulting application, moving users onto a supported release without losing established workflows.
- **Scientific systems joined up.** Integrated BIOVIA into the Oracle applications through BIOVIA workflow management, closing a manual handoff between laboratory and business systems.
- **Data flowed in automatically.** Designed the ETL pulling data into the Billing Portfolio from multiple source systems.
- **Performance engineered in.** Tuned SQL and PL/SQL with bulk processing so billing runs completed reliably, and automated supporting operations in advanced shell script.

Lead Technical Consultant · Telecommunications billing systems, Oracle 11g and 12c

Database design authority and production owner for telecom billing, and the database partner to a large Java engineering group.

- **New applications on solid ground.** Provided design and ongoing support for new telecommunications billing applications, setting the database patterns they were built on.
- **Billing kept running.** Maintained and monitored production billing applications, holding service steady on a revenue-critical path.
- **Java teams unblocked.** Acted as database solution partner to a large group of Java developers, converting application requirements into performant packages, procedures and functions.
- **Platform debt cleared.** Migrated the remaining Open VMS scripts into the Unix environment, retiring a platform dependency the team had carried for years.
- **Faster where it counted.** Applied advanced PL/SQL and bulk processing tuning to the heaviest billing workloads.

CSX Florida

Feb 2014 - Oct 2015

Lead Technical Consultant · Rail transport systems, Oracle 12c

Design and tuning authority for rail systems, with domain depth in transport operations and a standardized approach to database housekeeping.

- **Design that carried the domain.** Provided design and support for new rail system applications, backed by working expertise in railway transport operations.
- **Bottlenecks found and removed.** Diagnosed and resolved performance problems through query optimization and ADDM and AWR analysis, alongside application, memory and disk tuning.
- **Standard practice, everywhere.** Wrote purge scripts for trace files, alert logs and network logs that were adopted as the standard across every database in the estate.
- **Routine work scheduled away.** Implemented Unix cron and Windows Task Scheduler automation for day-to-day activities, and kept technical documentation current as systems changed.
- **Problems seen early.** Made trace, alert, metric and log examination routine at both database and grid infrastructure level, catching issues before users reported them.

Infogroup USA

Nov 2012 - Feb 2014

Lead Technical Consultant · Marketing solutions, Oracle 11g

Technical lead for product implementation and automation, with performance, security and monitoring of the marketing data estate.

- **Implementation and automation delivered.** Provided technical services for product implementation and built the advanced Unix shell automation supporting it.
- **Resource contention resolved.** Monitored CPU, I/O contention and memory paging continuously to identify bottlenecks, then reorganized to hold performance as demand grew.
- **Worst queries fixed first.** Targeted the top SQL by response time and disk reads per execution and tuned them using explain plan, TKPROF and SQL Tuning Advisor.
- **Storage made to perform.** Defragmented tablespace data blocks and partitioned large tables, recovering performance without additional hardware.
- **Refreshes without hands.** Automated database refresh from the host system into Oracle through SQL and PL/SQL procedures, and resized online redo logs across production and standby.
- **Estate kept secure and visible.** Owned user, space and security management, maintained database integrity and access control, and reported status through daily and nightly production reporting.

First Data Corporation USA

Jul 2012 - Nov 2012

Lead Technical Consultant · Open Loop card platform, Oracle 11g

Production stability and scheduled processing for a credit card platform.

- **Critical issues resolved.** Supported the Open Loop project through critical production incidents, restoring service on a finance and credit card platform.
- **Scheduled work made dependable.** Managed Oracle jobs through the CA scheduler, giving batch processing a predictable operating rhythm.
- **Logic hardened.** Wrote and tuned the triggers, packages, procedures and functions the platform depended on, applied with domain knowledge of finance and credit card processing.

MBIA USA

Jul 2010 - Jul 2012

Lead Technical Consultant · Loan Putback System, financial guarantee insurance

Designed and delivered a system central to the firm's loan repurchase recovery, and the reporting layer executives used to act on it.

- **System that recovered value.** Designed and delivered Phase I of the Loan Putback System, tracking loans where representations or warranties were breached so that offending loans could be put back to the sponsor for repurchase at outstanding principal plus accrued interest.
- **Architecture owned.** Led the database architecture and advanced PL/SQL behind the system, applying domain knowledge of finance and insurance to the deployment strategy.
- **Reporting leadership could use.** Built advanced reporting in SSRS 2008 and management reporting through OBIEE and BI Publisher, putting portfolio position in front of decision makers.
- **Critical applications held steady.** Provided production support and query tuning across MBIA's critical applications.

British Petroleum UAE

Jun 2005 - Apr 2010

Lead Technical Consultant · Oracle PL/SQL, Pro*C and shell, E-Business Suite

Solution architect and technical lead across Oracle applications, platform migration and enterprise integration, and the mentor the delivery teams learned from.

- **Migrations completed at scale.** Architected and migrated a large Oracle custom application to Oracle 10g, and carried the MIT Boston VAX-based Oracle application onto Unix, retiring end-of-life platforms.
- **Platform dependency removed.** Converted VMS environment scripts into Unix Korn shell as part of the server migration, leaving a single supportable scripting standard.
- **Enterprise systems integrated.** Delivered the Microsoft Web Portal and JD Edwards integration, building the components that pulled JD Edwards data into the portal.
- **Reuse institutionalized.** Built a repository of reusable code and project documentation that later projects drew on instead of rebuilding.
- **Quality through review.** Held the lead technical role on code reviews and design decisions, and mentored teams on implementation strategy, integration and best practice.
- **Client capability raised.** Ran Oracle SQL and PL/SQL training for client teams and supported pre-sales with proof-of-concept builds that won work.

Earlier experience

ICICI Infotech India · **Lead Technical Consultant**

2004 - 2005

Implemented the Orion ERP product for US and Canada clients including Custom Glass Products and Pliva from offshore, and architected the onsite and offshore coordination model along with the RFP, estimation and costing practice behind it.

Sundaram Infotech India · **Associate Lead Technical Consultant**

2003 - 2004

Delivered the LAMS non-banking finance product covering hire purchase, leasing, bill discounting and fixed deposits for Mahindra & Mahindra Finance, and migrated Cronos UK's Unix-based Oracle application onto a Java platform.

Focus Infotech USA · **Senior Project Leader**

1999 - 2003

Delivered supply chain, portfolio management, payroll and HRMS and insurance systems for SoftBrands Hospitality, DataForce Systems and Vadum, established the offshore delivery model and onsite communication strategy, and completed a large Oracle reports migration.

Unraven Solutions USA · **Systems Analyst**

1998 - 1999

Converted Galileo's character-based airline ticketing system in Denver to a GUI application, designing the reusable code approach the conversion ran on.

3i Solutions

India

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Senior Systems Executive

1995 - 1997

Built inventory, sales forecasting and chargeback management systems for Standard Chartered Bank across India, Malaysia and Thailand, and for Hindustan Lever, owning database design and review.

Sundaram Information Systems

India

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Software Developer

1994 - 1995

Delivered MODVAT, export invoicing and export freight systems, and the PIONEER hire purchase, leasing, bill discounting and fixed deposit finance modules for leasing and investment clients.

Global Solutions

India

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Software Developer

1993 - 1994

Built material management, inventory monitoring, order processing and financial accounting systems for a portfolio of manufacturing and engineering clients.

Domain expertise

Healthcare payment integrity

Banking & financial systems

Credit card & dispute management

Animal health

Telecommunications billing

Rail transport

Energy & petroleum

Supply chain management

Marketing solutions

Insurance

Hire purchase, leasing & bill discounting

Fixed deposits & financial accounting

Clients served · Galileo, Vadum, DataForce Systems, SoftBrands Hospitality, Cronos (UK), Baker Haus Veit (Canada), Custom Glass Products, Pliva, Massachusetts Institute of Technology, British Petroleum (UAE), Standard Chartered Bank (India, Malaysia and Thailand), Hindustan Lever, ICARDA (Syria), Electricity Holding Company (Oman), Diwan of Royal Court (Oman), Bank Sohar (Oman), Oman Telecommunications

References available on request.